

the fact that the two sets of intermediates were not structurally related to each other since the orientation of the intermediates towards the end products permitted the individual technical problems addressed by the intermediates to be combined into a unitary overall problem to the solution of which both sets of purpose-made intermediates contributed.

In W 35/91 the board ruled that the requirements of unity of invention as set out in R. 13.1 and R. 13.2 PCT were met if the novel intermediates designed to give rise to the novel end products were technically sufficiently closely related by their contribution to an essential structural element of the end products.

In W 7/85 (OJ 1988, 211) the board stated that there was sufficient technical information to justify a prima facie finding of unity between a claim to a mixture and a claim to one essential component of that mixture or a narrowly defined version thereof. If a finding of unity was justified in cases of chemical intermediates and end products even when, as was often the case, only a part of the intermediate structure was actually incorporated, there was all the more reason to view the intact components and the corresponding compositions in a mixture as technically interconnected by incorporation. The former were not even destroyed when the admixture was prepared and fully retained their properties and functions in the product, unlike typical intermediates, which lost their identity in the process. Thus, both inventions could be considered to fall within the same general inventive concept. In such cases the requirement that the means for preparing the end product should be "specially designed for carrying out the process" appeared to be fulfilled since none of the means led or was related to an end product outside the scope of its definition. In view of this the character of the invention in the component was, prima facie, also dependent on the existence of an invention in the end product.

3. Assessing lack of unity of invention

3.1. General approach – content of claims

In W 6/97 it was held that the determination of unity of invention must be made on the basis of the contents of the claims as interpreted in the light of the description and of the drawings, if applicable (with reference to Annex B, Part 1(b) PCT Administrative Instructions; cf. now Annex B, para. (b) PCT Administrative Instructions, as in force from 1.1.2022).

In W 39/90 the board observed that it was not the formal choice of words or form of reference, but the actual content of the claims which established technical relationships between the subject-matter of different claims, and which was thus decisive for the question of unity. In W 33/92 the board emphasised that R. 13.1 PCT did not require the link between the subject-matter of the two independent claims to be expressly stated in their wording. All that was required was that there be a single inventive concept.

3.2. Assessment of lack of unity at the search stage

Lack of unity may be directly evident **a priori**, i.e. before examination of the merits of the claims in comparison with the state of the art revealed by the search (W 1/96 and W 6/90,

OJ 1991, 438). If the objection of lack of unity is raised a priori, the technical problem must be defined solely on the basis of the description and not of the prior art (see W 50/91, W 52/91, W 22/92, W 52/92 and T 188/04).

The powers of the EPO as ISA with regard to the assessment of unity **a posteriori**, i.e. after taking into account the prior art brought to light by the search, were clarified by the Enlarged Board of Appeal in G 1/89 (OJ 1991, 155 – referral by a board of appeal in W 12/89, OJ 1990, 152) and G 2/89 (OJ 1991, 166 – referral by the President of the EPO). The referring board asked the Enlarged Board inter alia if an ISA has the power to carry out a substantive examination of an international application in respect of novelty and inventive step when considering under Art. 17(3)(a) PCT whether the application complies with the requirement of unity of invention set forth in R. 13.1 PCT. The President of the EPO requested the Enlarged Board's opinion on whether the EPO as ISA may request a further search fee where the international application is considered to lack unity of invention a posteriori. The referred points of law were dealt with in consolidated proceedings.

The Enlarged Board noted that the procedural separation of search and (substantive) examination under the PCT, as well as under the EPC 1973, led to a certain overlapping because of the functional relationship between search and examination. Thus, although the objective of the search was in principle limited to discovering and reporting on relevant prior art for the purpose of assessing novelty and inventive step, which was ultimately the task of the examining authority (i.e. the IPEA and/or the designated Office under the PCT and the examining division under the EPC 1973), in many instances the search examiner needed to form a provisional opinion on these issues in order to carry out an effective search (for more on this aspect see also the summary in this chapter II.B.4.2.).

The Enlarged Board further noted that neither in the PCT nor in the Rules under the PCT were there provisions indicating how to decide whether an international application complies with the requirement of unity of invention. However, the PCT Search Guidelines stated that lack of unity may be directly evident "a priori", or only become apparent "a posteriori" (see now para. 10.03 PCT International Search and Preliminary Examination Guidelines as in force from 1.3.2022). They contained a direct reference to the consideration of unity by the ISA on an a posteriori basis, i.e. after an assessment of the claims with regard to novelty and inventive step in relation to the prior art. The corresponding EPO Guidelines showed that under the EPC 1973 it was also clearly foreseen that consideration of unity by the search divisions could be carried out on an a posteriori basis (i.e. formerly Guidelines, B-VII.5; cf. R. 46 EPC 1973; see now Guidelines – March 2022 version, in particular F-V.4, and R. 64 EPC). Therefore, the Guidelines had on this point to be considered as consistent with the PCT and the EPC 1973.

The Enlarged Board confirmed that the EPO when acting as an ISA and the boards of appeal when deciding on protests under Art. 17(3)(a) PCT were under an obligation to be guided by the PCT Search Guidelines (cf. Art. 2 of the then applicable agreement between the EPOrg and WIPO). Thus, in G 1/89 the Enlarged Board held that, as foreseen in these guidelines, an international application may, under Art. 17(3)(a) PCT, be considered not to comply with the requirement of unity of invention, not only "a priori" but also "a

posteriori". According to G. 2/89, the EPO in its function as an ISA may, pursuant to Art. 17(3)(a) PCT, request a further search fee where the international application is considered to lack unity of invention "a posteriori". The principle whereby the EPO may also find a lack of unity "a posteriori" when carrying out an international search (G. 1/89, OJ 1991, 155; G. 2/89, OJ 1991, 166) applies equally to a European search because in both cases the searches and search reports are virtually identical (see T. 87/88, OJ 1993, 430).

According to T. 94/91, Art. 82 EPC 1973 clearly provided that European patent applications must satisfy the criteria of unity of invention without any restriction. There was no difference in the EPC 1973 between "a priori" and "a posteriori" lack of unity. It was of no importance whether the non-unity appeared immediately or only in view of a document found during the search or the examination. This was also the interpretation given in the Guidelines and in G. 2/89 (OJ 1991, 166), which, although it related to PCT applications, concerned a similar procedure.

Where the search division makes a finding of non-unity, the applicant is normally informed, along with the partial search report relating to the invention, or group of inventions, first mentioned in the claims, that a further search fee must be paid for the other inventions which it wishes to be covered by the European search report (R. 64(1) EPC). Art. 17(3)(a) PCT lays down the equivalent procedure before the ISA. R. 164 EPC contains special provisions for Euro-PCT applications having entered the European phase. Relevant decisions are summarised in this chapter II.B.6. "Plurality of inventions – further search fees".

3.3. Assessment of lack of unity and of requests for refund of further search fees in examination proceedings

The requirement of unity under Art. 82 EPC has to be satisfied not only by the patent application as filed but also at later stages of the patent granting procedure, i.e. before the examining division, up until grant of the patent. Under R. 64(2) EPC (R. 46(2) EPC 1973) the examining division may disagree with the search division's finding of lack of unity, and the examining division's opinion is decisive in that the latter may order the refund of any further search fee at the applicant's request (see cases below and in this chapter II.B.6). It lies within the discretion of the examining division to decide that a patent application lacks unity of invention within the meaning of Art. 82 EPC, even if the search division did not raise a similar objection (T. 178/84, OJ 1989, 157). See also T. 631/97 (OJ 2001, 13), summarised in this chapter II.B.6.1. "Consequences of non-payment of a further search fee".

In T. 544/88 (OJ 1990, 429) the board ruled that if an applicant, in response to an objection of lack of unity, filed new claims allegedly relating to an invention meeting the unity requirement, examination should be continued even if the other application documents had not been limited to the subject-matter of those claims. However, the examining division could request the applicant to bring the description and drawings into line with the valid claims (R. 27(1)(d) EPC 1973 as in force until 31.5.1991; cf. subsequent R. 27(1)(c) EPC 1973, now R. 42(1)(c) EPC) and to delete those parts of the description

and drawings which did not refer to claimed inventions (R. 34(1)(c) EPC 1973). It would then be necessary to check in each case whether such adjustments ought to be deferred until allowable claims were submitted.

In J 24/96 (OJ 2001, 434) the board held that within the framework of R. 46 EPC 1973 it was the task of the examining divisions (and the boards of appeal) to examine whether communications of the search divisions under R. 46(1) EPC 1973 asking for further search fees were justified. R. 46(2) EPC 1973 did not, however, refer to any acts of the search divisions other than communications issued under R. 46(1) EPC 1973. It was therefore not necessary for an examining division to deal in a decision under R. 46(2) EPC 1973 with other objections raised by the applicants in connection with the search, such as that the partial search report was not drawn up in accordance with the Guidelines. The board observed that the examining divisions of course have the possibility of arranging for an additional search to be performed, if they consider a search report delivered by the search division incomplete, but that this had nothing to do with a formal decision issued under R. 46(2) EPC 1973 concerning the refund of further search fees upon request.

In T 188/00 the board pointed out that the review by the examining division under R. 46(2) EPC 1973 had to be carried out having regard only to the facts presented by the search division in its communication under R. 46(1) EPC 1973. Since in most cases objections against lack of unity were raised having regard to the prior art (i.e. a posteriori), the examining division had to base its review solely on the documents cited in the partial search report and on the specification of the different inventions drawn up by the search division, while taking into account any arguments of the applicant in support of a request for a refund. Concerning the analogous protest procedures under the PCT (R. 40.2(c) and (e) and R. 68.3(c) and (e) PCT), the boards of appeal had ruled that these reviews had to be based exclusively on the reasons given in the invitation to pay additional fees, having regard to the facts and arguments submitted by the applicants (W 4/93, OJ 1994, 939). However, in the case in hand the examining division had reasoned that the claims as filed lacked unity a posteriori as a consequence of the finding that claim 1 as filed lacked an inventive step having regard inter alia to document D4, was not cited in the partial search report. Thus the examining division had not reviewed the finding of lack of unity of invention within the meaning of R. 46(2) EPC 1973, but had carried out a fresh examination on the basis of a new document. Therefore its decision to refuse the refund of a further search fee had to be set aside for formal reasons alone; the appellant's refund request was justified. See also T 1414/18, with further references; also T 2873/19, in which the refund request was refused.

In the order of its decision on a requested refund of (further) search fees, the examining division must clearly specify the request which has been decided on, the extent to which it grants that request and orders a refund, and the outcome for any remaining parts of the request. Allowing a separate appeal against its interlocutory decision under Art. 106(2) EPC is a constitutive decision of the examining division which establishes the possibility of challenging the interlocutory decision by way of an appeal, and therefore it too should be included in the order (T 756/14).

3.4. No assessment of lack of unity in opposition proceedings

In G 1/91 (OJ 1992, 253) the Enlarged Board held that unity of invention did not come under the requirements which a European patent and the invention to which it related had to meet under Art. 102(3) EPC 1973 (Art. 101(3)(a) EPC) when the patent was maintained in amended form. It was consequently irrelevant in opposition proceedings that the European patent as granted or amended did not meet the requirement of unity. It was clear from the "travaux préparatoires" not only that lack of unity was deliberately excluded as a ground for opposition, but also that any lack of unity arising during opposition proceedings as a result of amendment of the patent had to be accepted. Although unity of invention under Art. 82 EPC 1973 was a material requirement, it was still merely an administrative regulation. It served a number of administrative purposes, particularly in demarcating the respective responsibilities of the departments. The administrative purposes of unity were fulfilled in the main up to the time the patent was granted. The purpose of opposition proceedings was to give a competitor the opportunity of opposing unjustified protective rights. He did not also need to be given the opportunity of contesting a patent on the ground of lack of unity. Lack of unity did not in fact rule out patent protection; it could only result in an application being divided to produce two or more patents. See also T 830/11.

4. Criteria for determining lack of unity

4.1. Analysis of the technical problem as a precondition

According to the established case law of the boards of appeal, determining unity of invention requires as a precondition an analysis of the technical problem or problems underlying the respective group of inventions (see W 11/89, OJ 1993, 225; W 6/97, T 188/04), i.e. whether or not the subject-matter claimed as the solution to such a problem represents a single general inventive concept (see W 6/91; also T 2248/12 and T 129/14). Disregarding this principle would be in itself sufficient justification for the reimbursement of the additional search fees. In W 8/94 the board held that a discussion of the problem underlying the claimed subject-matter was required, because only then was it possible to decide whether or not a common special technical feature within the meaning of R. 13.1 and R. 13.2 PCT existed for different embodiments (see W 11/89, OJ 1993, 225; W 14/89, W 59/90, W 14/91, W 17/91; also T 1414/18, with further references).

In W 6/97 the board found that establishing the technical problem underlying a claimed invention or group of inventions in relation to the state of the art should start, as a rule, from what was considered in the description as having been achieved by the claimed invention, since claims directed to compositions of matter at least were normally silent on the technical effects to be achieved by such compositions. As soon as the search revealed prior art which was clearly more relevant than that already acknowledged in the description of the international application, it was necessary to determine what was to be considered as the particular technical problem in view of both the disclosure of the international application as a whole and the prior art thus revealed (see W 6/91). Unity of invention might be assessed only after the technical problem had been determined in such a manner.